

Noise Reduction in Superconducting Qubits Using RF Switch



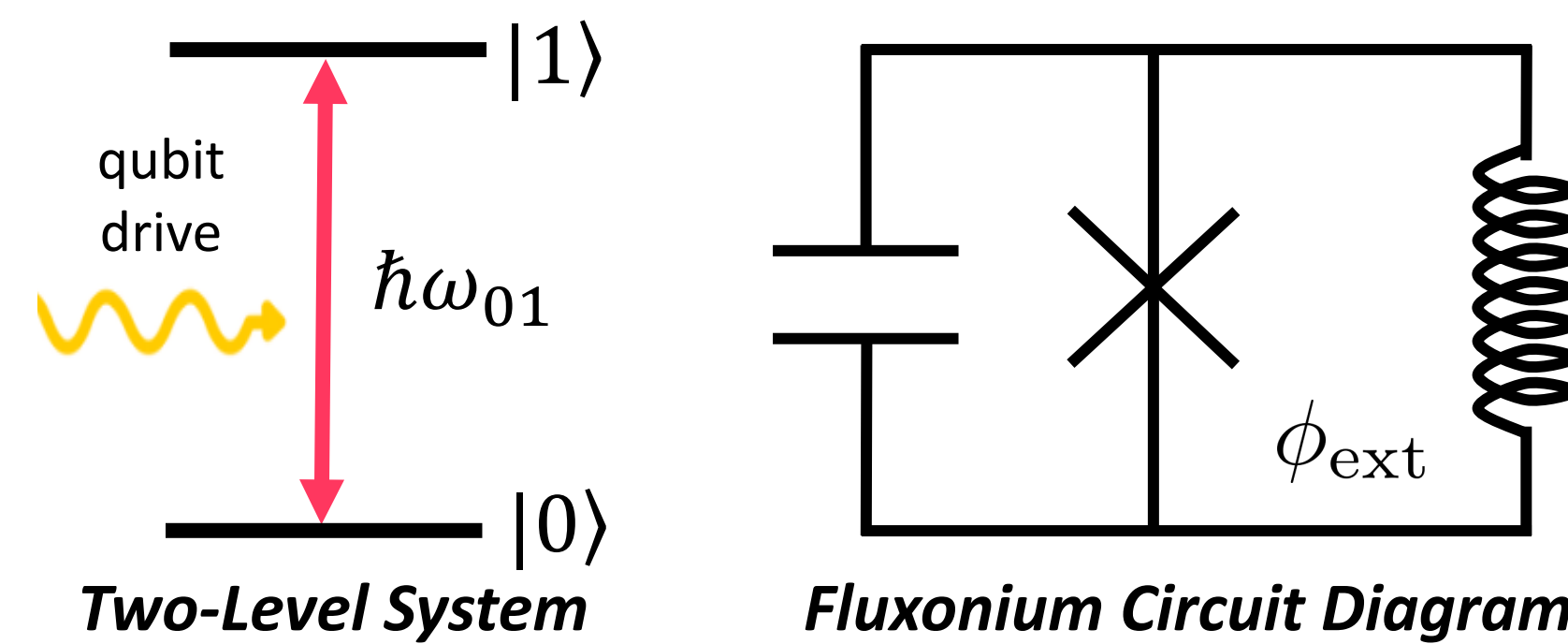
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Background

Superconducting Qubits

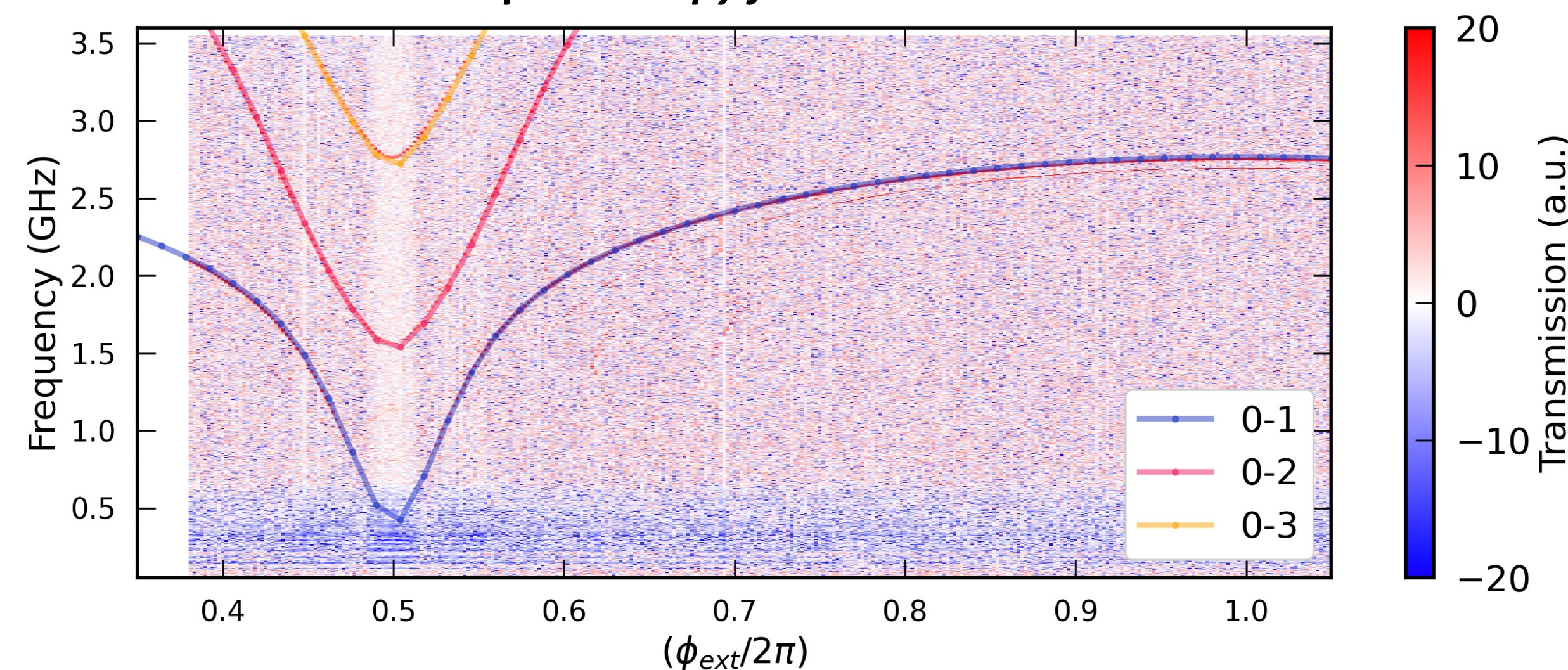
- ❖ Superconducting qubits: “artificial atom”
- ❖ Fluxonium: flux-tunable qubit with rich energy level structure



Hamiltonian for Fluxonium [1]

$$H = 4E_C n^2 - E_J \cos(\phi + \phi_{ext}) + \frac{1}{2} E_L \phi^2$$

Two-Tone Spectroscopy for Tested Fluxonium Device



Motivation

- ❖ Fluxonium can have small matrix elements → harder to drive qubit transitions → requires amplification
- ❖ Amplifiers introduce noise → reduce qubit single shot fidelity and coherence time

Project Goal

- ❖ Implement a switch to gate qubit pulses → reduce noise while not driving qubit → improve qubit single shot fidelity

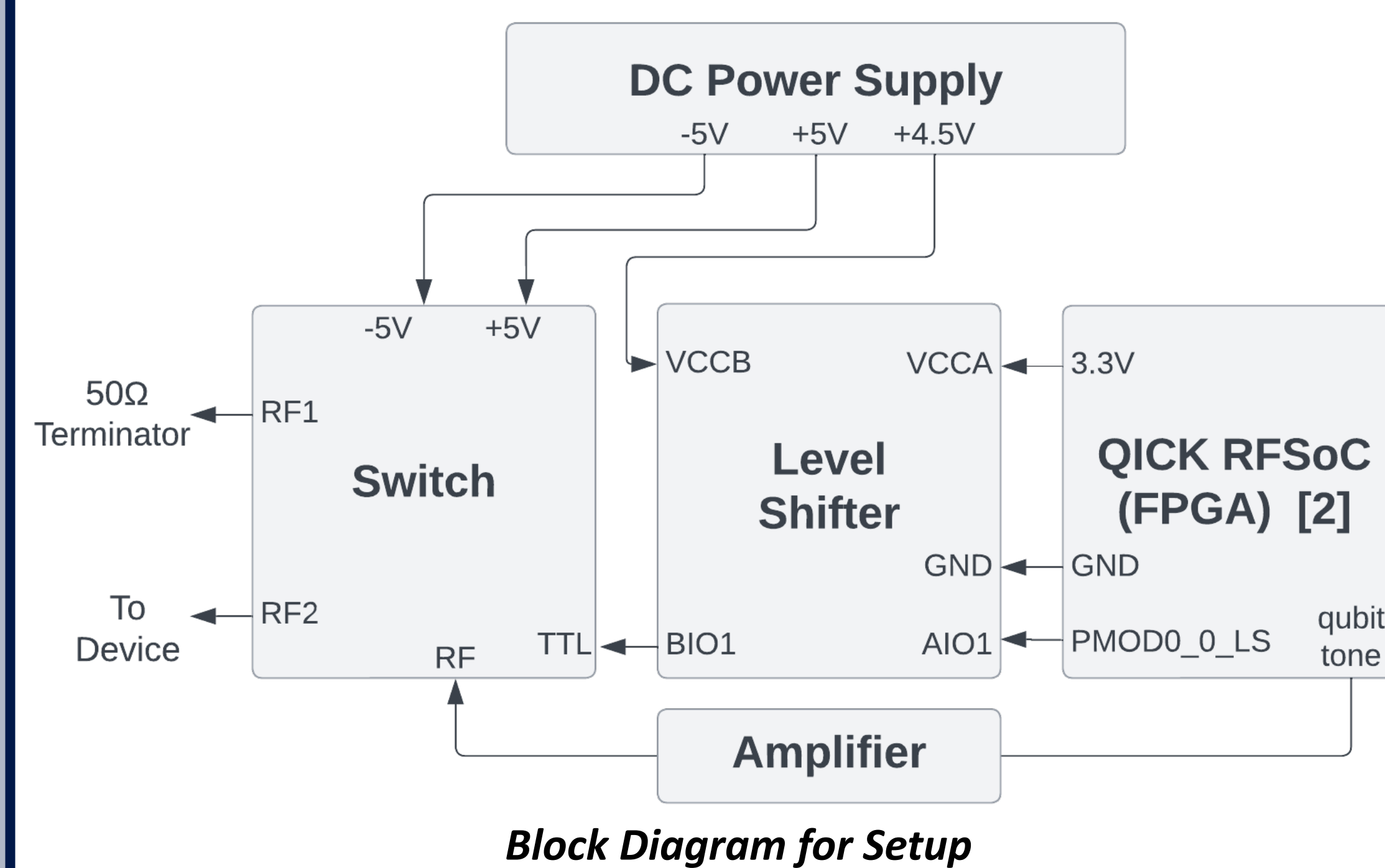
Qubit Pulse Without Switch



Qubit Pulse With Switch



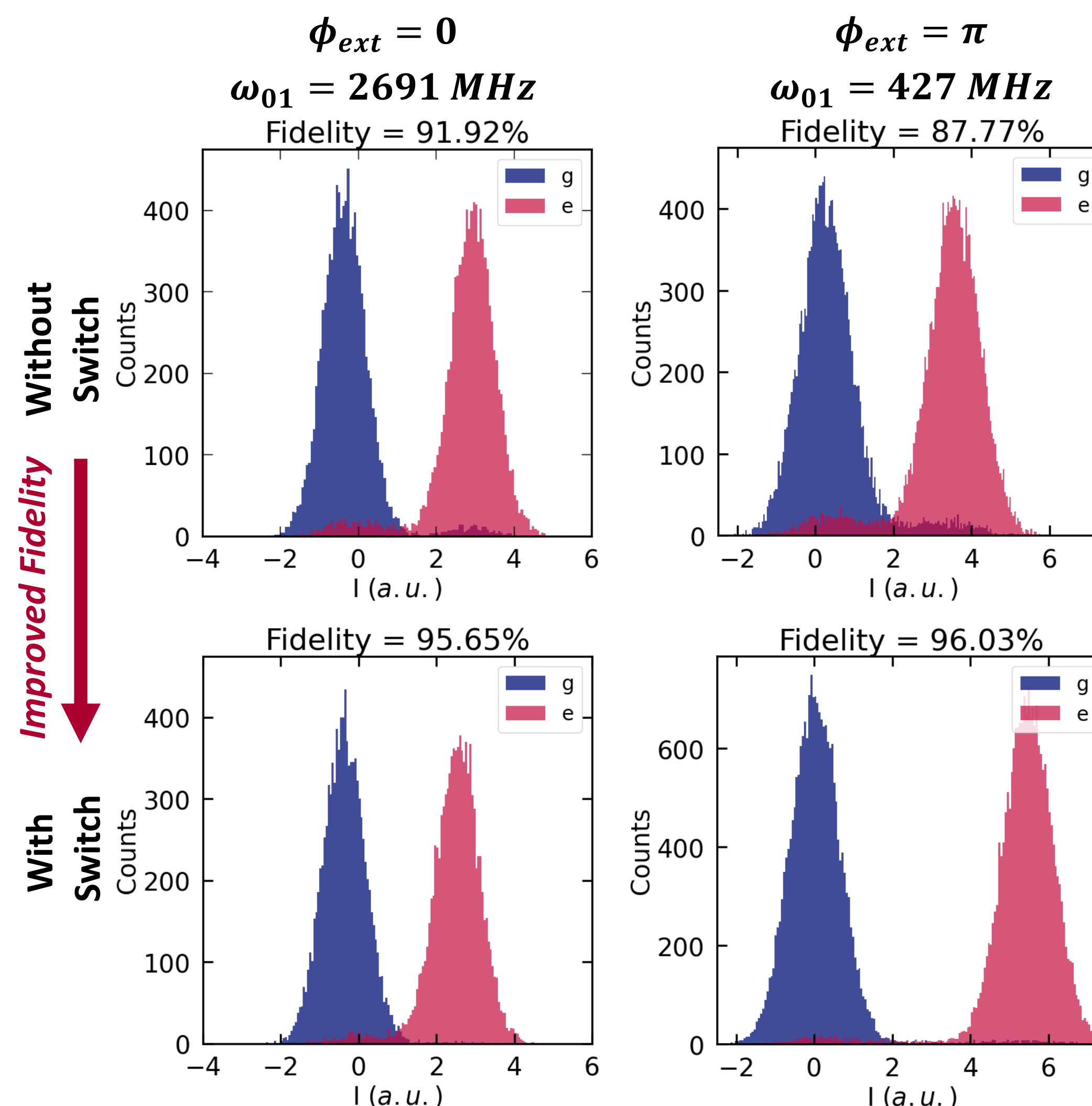
Methods



Results

Qubit Single Shot Fidelity

Comparison of Qubit Single Shot Fidelity Without and With Switch

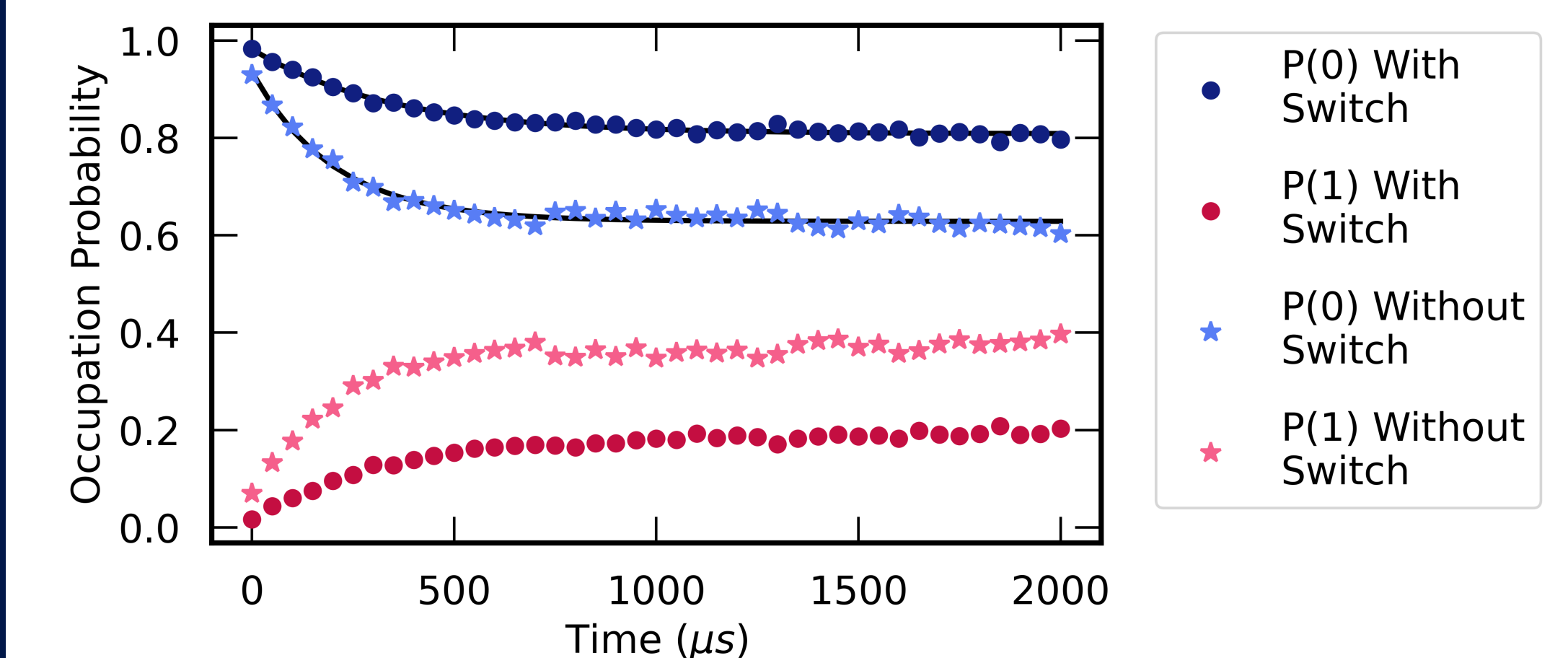


- ❖ Used post selection for data at $\phi_{ext} = \pi$
- ❖ Fidelity calculated via $F = 1 - P(e|g) - P(g|e)$ [3]

Results (cont.)

State Lifetime

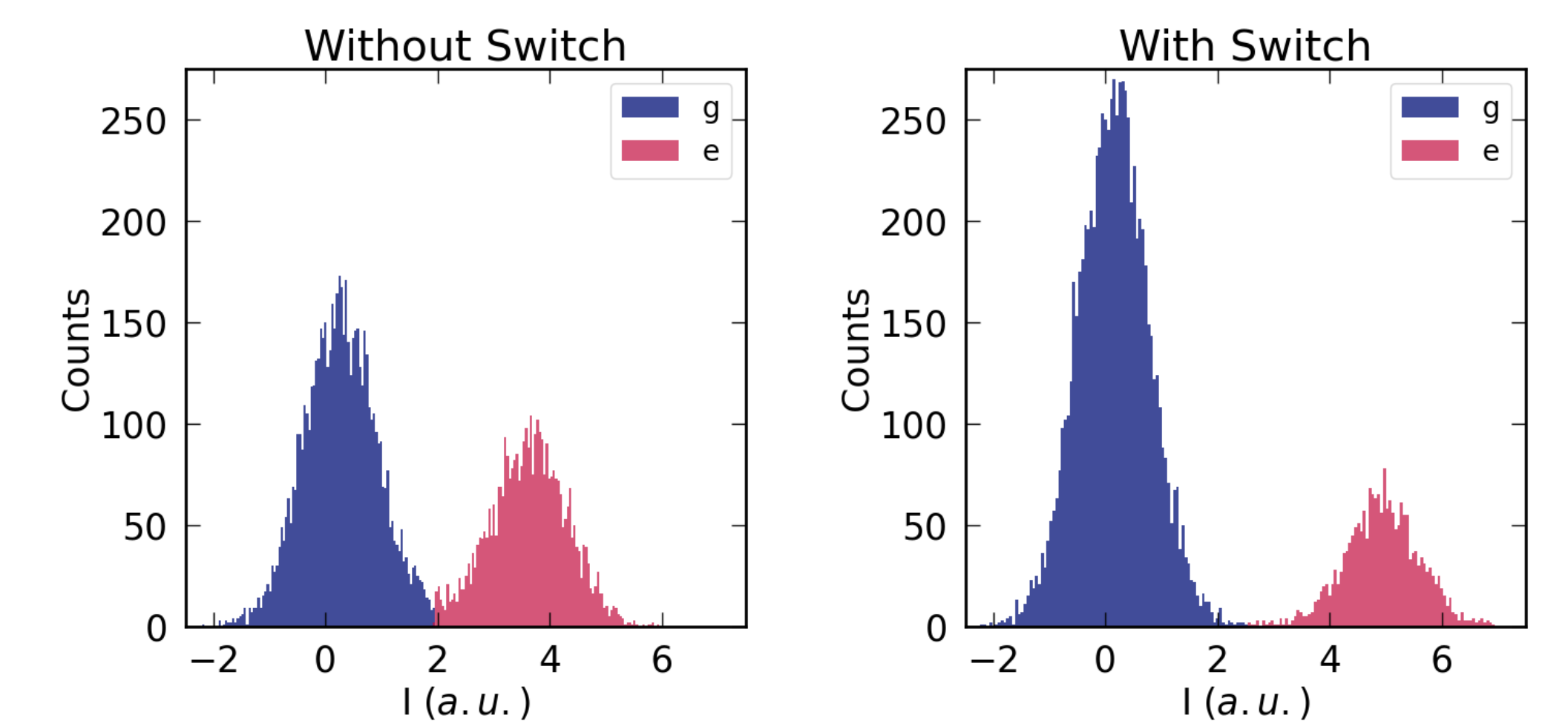
Comparison of Ground State Lifetimes
Without and With Switch, $\phi_{ext} = \pi$



Qubit Lifetime Measurement Results

	T_1 (μs)	$P_f(0)$	Effective Qubit Temperature (mK)
Without Switch	201 ± 11	62%	~ 30
With Switch	342 ± 16	80%	~ 14

Thermalized State Populations Without and With Switch



Future Work

- ❖ Characterize switch loss and behavior at higher frequencies
- ❖ Integrate code with experiments requiring multiple qubit pulses (e.g.: Ramsey, Hahn Echo, etc.)

References

- [1] V. E. Manucharyan et al., *Science* 326, 113–116 (2009).
- [2] T. Stefanazzi et al., *Rev. Sci. Instrum.*, 93, 044709 (2022).
- [3] T. Walter et al., *Phys. Rev. Applied* 7, 054020 (2017).